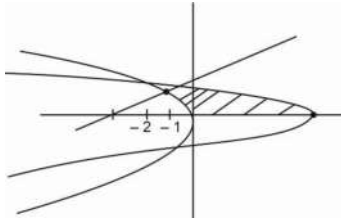


181. Let the area of the region $\{(x, y) : x - 2y + 4 \geq 0, x + 2y^2 \geq 0, x + 4y^2 \leq 8, y \geq 0\}$ be $\frac{m}{n}$, where m and n are coprime numbers. Then $m + n$ is equal to _____.

माना क्षेत्र $\{(x, y) : x - 2y + 4 \geq 0, x + 2y^2 \geq 0, x + 4y^2 \leq 8, y \geq 0\}$ का क्षेत्रफल $\frac{m}{n}$ है, जहाँ m तथा n असहभाज्य संख्याएँ हैं। तो $m + n$ बराबर है _____।

Ans. 119

Sol.



Area between these curve will be

$$\int_{-2}^{-1} \left(-\sqrt{\frac{-x}{2}} + \frac{x+4}{2} \right) dx + \int_{-1}^0 \left(\sqrt{\frac{8-x}{4}} - \sqrt{\frac{-x}{2}} \right) dx + \int_0^8 \sqrt{\frac{8-x}{4}} dx = \frac{107}{12} = \frac{m}{n}$$

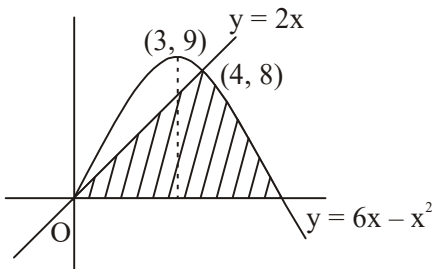
$$\Rightarrow m + n = 119$$

182. If the area of the region $\{(x, y) : 0 \leq y \leq \min\{2x, 6x - x^2\}\}$ is A , then $12A$ is equal to _____.

यदि क्षेत्र $\{(x, y) : 0 \leq y \leq \min\{2x, 6x - x^2\}\}$ का क्षेत्रफल A है, तो $12A$ बराबर है _____

Ans. 304

Sol.



Solving $y = 2x$ and $y = 6x - x^2$

$$2x = 6x - x^2$$

$$x^2 - 4x = 0$$

$$y = x = 0, x = 4, y = 8$$

$$A = \frac{1}{2} 4.8 + \int_4^6 (6x - x^2) dx$$

$$= 16 + \left(3x^2 - \frac{x^3}{3} \right)_4^6$$

$$A = 16 + 3(36 - 16) - \frac{1}{3}(216 - 64)$$

$$A = 16 + 60 - \frac{152}{3}$$

$$A = 16 + \frac{28}{3}$$

$$12A = 304$$

183. The area (in sq. units) of the part of the circle $x^2 + y^2 = 169$ which is below the line $5x - y = 13$ is

$$\frac{\pi\alpha}{2\beta} - \frac{65}{2} + \frac{\alpha}{\beta} \sin^{-1}\left(\frac{12}{13}\right), \text{ where } \alpha, \beta \text{ are coprime number. Then } \alpha + \beta \text{ is equal to } \underline{\hspace{2cm}}.$$

वृत्त $x^2 + y^2 = 169$ के रेखा $5x - y = 13$ से नीचे के भाग का क्षेत्रफल (वर्ग इकाई में) $\frac{\pi\alpha}{2\beta} - \frac{65}{2} + \frac{\alpha}{\beta} \sin^{-1}\left(\frac{12}{13}\right)$ है, जहाँ α , β असहभाज्य संख्याएँ हैं। तो $\alpha + \beta$ बराबर है $\underline{\hspace{2cm}}$.

Ans. 171

Sol. Solving line and circle $x = 0, 5$

$$\tan \alpha = \frac{12}{5}$$

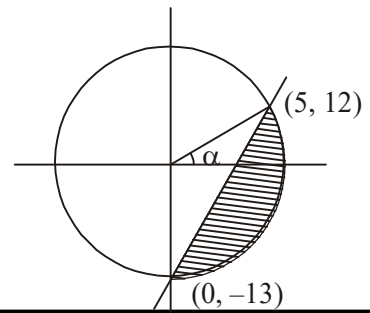
$$\sin \alpha = \frac{12}{13}$$

$$\sin\left(\frac{\pi}{2} + \alpha\right) = \cos \alpha = \frac{5}{13}$$

$$\text{Now area of segment} = \frac{1}{2}(r^2)(\theta - \sin \theta)$$

$$= \frac{1}{2} \times 169 \times \left[\frac{\pi}{2} + \sin^{-1} \alpha - \sin\left(\frac{\pi}{2} + \alpha\right) \right]$$

$$= \frac{169}{2} \left[\frac{\pi}{2} + \sin^{-1} \frac{12}{13} - \frac{5}{13} \right]$$



$$= \frac{169\pi}{4} + \frac{169}{2} \sin^{-1} \frac{12}{13} - \frac{65}{2}$$

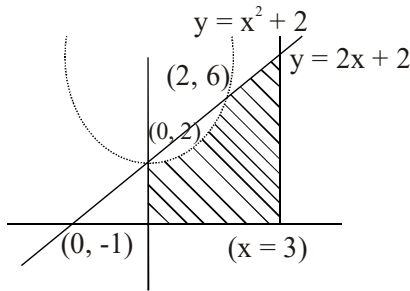
$$\alpha = 169, \beta = 2$$

184. Let the area of the region $\{(x, y) : 0 \leq x \leq 3, 0 \leq y \leq \min \{x^2 + 2, 2x + 2\}\}$ be A. Then 12A is equal to :

माना क्षेत्र $\{(x, y) : 0 \leq x \leq 3, 0 \leq y \leq \min \{x^2 + 2, 2x + 2\}\}$ का क्षेत्रफल A है। तो 12A बराबर है।

Ans. 164

Sol.



$$A = \int_0^2 (x^2 + 2) dx + \int_2^3 (2x + 2) dx$$

$$A = \left(\frac{x^3}{3} + 2x \right)_0^2 + (x^2 + 2x)_2^3$$

$$A = \frac{41}{3} \Rightarrow 12A = 164$$

185. Three points $O(0, 0)$, $P(a, a^2)$, $Q(-b, b^2)$, $a > 0, b > 0$, are on the parabola $y = x^2$. Let S_1 be the area of the region bounded by the line PQ and the parabola, and S_2 be the area of the triangle OPQ. If the minimum value of $\frac{S_1}{S_2}$ is

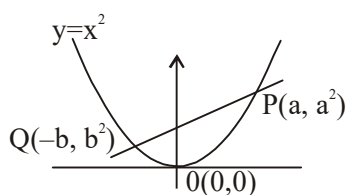
$\frac{m}{n}$, $\gcd(m, n) = 1$, then $m + n$ is equal to _____.

परवलय $y = x^2$ पर तीन बिंदु $O(0, 0)$, $P(a, a^2)$, $Q(-b, b^2)$, $a > 0, b > 0$ हैं। माना रेखा PQ तथा परवलय से घिरे क्षेत्र का क्षेत्रफल

S_1 है तथा त्रिभुज OPQ का क्षेत्रफल S_2 है। यदि $\frac{S_1}{S_2}$ का न्यूनतम मान $\frac{m}{n}$, $\gcd(m, n) = 1$ हैं, तो $m + n$ बराबर है _____

Ans. 7

Sol.



Equation of PQ

$$y + b = \frac{a^2 - b^2}{a + b}(x - b^2)$$

$$\Rightarrow y = (a - b)x + ab$$

$$S_1 = \int_{-b}^a ((a - b)x + ab - x^2) dx$$

$$= \left(\frac{(a - b)}{2} x^2 + abx - \frac{x^3}{3} \right)_{-b}^a$$

$$= \frac{1}{6}(a + b)^3$$

$$S_2 = \frac{1}{2} \begin{vmatrix} -b & b^2 & 1 \\ 0 & 0 & 1 \\ a & a^2 & 1 \end{vmatrix} = \frac{1}{2} ab(a + b)$$

$$\frac{S_1}{S_2} = \frac{\frac{1}{6}(a + b)^3}{\frac{1}{2} \cdot ab(a + b)}$$

$$= \frac{1}{3} \frac{(a + b)^2}{ab} = \frac{1}{3} \left(\frac{a}{b} + \frac{b}{a} + 2 \right)$$

$$\therefore \frac{b}{a} + \frac{a}{b} \geq 2$$

$$\Rightarrow \left(\frac{S_1}{S_2} \right)_{\min} = \frac{4}{3} = \frac{m}{n}$$

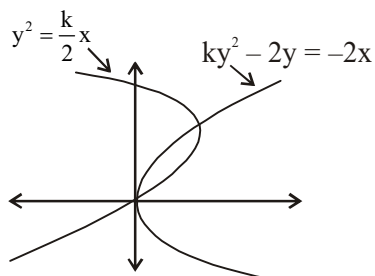
$$\Rightarrow m + n = 7$$

186. The sum of squares of all possible values of k , for which area of the region bounded by the parabolas $2y^2 = kx$ and $ky^2 = 2(y - x)$ is maximum, is equal to _____.

k के सभी संभव मानों, जिनके लिए परवलयों $2y^2 = kx$ तथा $ky^2 = 2(y - x)$ से घिरे क्षेत्र का क्षेत्रफल अधिकतम है, के वर्गों का योग है

Ans. 8

Sol.



$$\text{Area} = A = \int_0^{\frac{2k}{k^2+4}} \left[\left(\frac{ky^2 - 2y}{-2} \right) - \left(\frac{2y^2}{k} \right) \right] dy$$

$$\text{Area} = \left[\left(\frac{-k}{2} - \frac{2}{k} \right) \frac{y^3}{3} + \frac{y^2}{2} \right]_0^{\frac{2k}{k^2+4}}$$

$$\text{Area} = \frac{-(k^2 + 4)}{2k} \times \left(\frac{2k}{k^2 + 4} \right)^3 \times \frac{1}{3} + \left(\frac{2k}{k^2 + 4} \right)^2 \times \frac{1}{2}$$

$$= - \left(\frac{2k}{k^2 + 4} \right)^2 \times \frac{1}{3} + \left(\frac{2k}{k^2 + 4} \right)^2 \times \frac{1}{2}$$

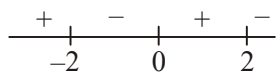
$$= \frac{1}{6} \times \frac{4k^2}{(k^2 + 4)^2}$$

$$A(k) = \frac{2}{3} \frac{k^2}{(k^2 + 4)^2}$$

$$A'(K) = \frac{2}{3} \frac{2K(K^2 + 4)^2 - 2K^2(K^2 + 4)(2K)}{(K^2 + 4)^4} = 0$$

$$\Rightarrow \frac{2}{3} (2K)(K^2 + 4) [K^2 + 4 - 2K^2] = 0$$

$$K = 0 \text{ or } K = 2 \text{ or } K = -2$$



$\therefore K = 0$ is local minima

$K = -2$ and 2 are local maxima

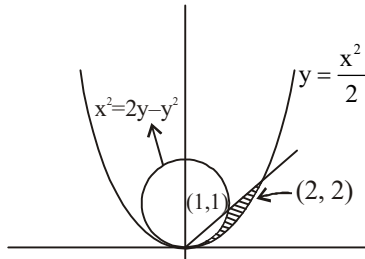
$$\therefore (-2)^2 + 2^2 = 8$$

187. If the area of the region $S = \{(x, y) : 2y - y^2 \leq x^2 \leq 2y, x \geq y\}$ is equal to $\frac{n+2}{n+1} - \frac{\pi}{n-1}$, then the natural number n is equal to _____.

यदि क्षेत्र $S = \{(x, y) : 2y - y^2 \leq x^2 \leq 2y, x \geq y\}$ का क्षेत्रफल $\frac{n+2}{n+1} - \frac{\pi}{n-1}$ के बराबर है, तब प्राकृतिक संख्या n बराबर है

Ans. 5

Sol. $2y - y^2 \leq x^2 \leq 2y, x \geq y$



$$\text{Area} = \int_1^2 (\sqrt{2y} - y) dy + \int_0^1 (\sqrt{2y} - \sqrt{2y - y^2}) dy$$

$$= \frac{7 - 2^{\frac{5}{2}}}{6} + \frac{2^{\frac{7}{2}} - 3\pi}{12}$$

$$= \frac{14 - 2^{\frac{7}{2}} + 2^{\frac{7}{2}} - 3\pi}{12} = \frac{7}{6} - \frac{\pi}{4}$$

$$n = 5$$

188. Let the area enclosed by the lines $x + y = 2$, $y = 0$, $x = 0$ and the curve $f(x) = \min \left\{ x^2 + \frac{3}{4}, 1 + [x] \right\}$ where $[x]$ denotes the greatest integer $\leq x$, be A . Then the value of $12A$ is _____.

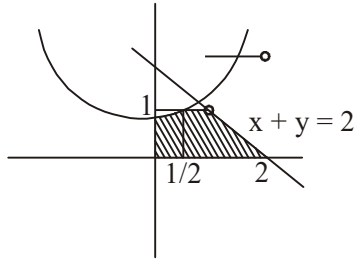
माना $[x]$ महत्तम पूर्णांक $\leq x$ है। माना रेखाओं $x + y = 2$, $y = 0$, $x = 0$ तथा वक्र $f(x) = \min \left\{ x^2 + \frac{3}{4}, 1 + [x] \right\}$ से घिरे क्षेत्र का

क्षेत्रफल A है। तो $12A$ का मान है :

Ans. 17

Sol. $A = \int_0^{\frac{1}{2}} \left(x^2 + \frac{3}{4} \right) dx + \frac{1}{2} \left(\frac{3}{2} + \frac{1}{2} \right) \times 2$

$$\frac{1}{3} \cdot \frac{1}{8} + \frac{3}{4} \cdot \frac{1}{2} + 1$$



$$12A = \frac{1}{2} + \frac{9}{2} + 12 = 17$$

189. Let $y = p(x)$ be the parabola passing through the points $(-1, 0)$, $(0, 1)$ and $(1, 0)$. If the area of the region $\{(x, y) : (x+1)^2 + (y-1)^2 \leq 1, y \leq p(x)\}$ is A , then $12(\pi - 4A)$ is equal to _____.

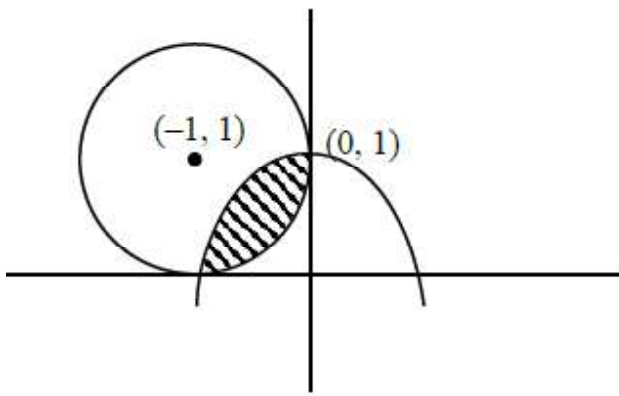
माना बिंदुओं $(-1, 0)$, $(0, 1)$ तथा $(1, 0)$ से होकर जाने वाला परवलय $y = p(x)$ है। यदि क्षेत्र

$\{(x, y) : (x+1)^2 + (y-1)^2 \leq 1, y \leq p(x)\}$ का क्षेत्रफल A है, तो $12(\pi - 4A)$ बराबर है _____

Ans. 16

Sol. There can be infinitely many parabolas through given points.

Let parabola $x^2 = -4a(y-1)$



Passes through $(1, 0)$

$$\therefore b = -4a(-1) \Rightarrow a = \frac{1}{4}$$

$$\therefore x^2 = -(y-1)$$

Now area covered by parabola $= \int_{-1}^0 (1-x^2)dx$

$$= \left(x - \frac{x^3}{3} \right)_1^0 = (0-0) - \left\{ -1 + \frac{1}{3} \right\}$$

$$= \frac{2}{3}$$

Required Area = Area of sector – {Area of square – Area covered by Parabola}

$$= \frac{\pi}{4} - \left\{ 1 - \frac{2}{3} \right\}$$

$$= \frac{\pi}{4} - \frac{1}{3}$$

$$\therefore 12(\pi - 4A) = 12 \left[\pi - 4 \left(\frac{\pi}{4} - \frac{1}{3} \right) \right]$$

$$= 12 \left[\pi - \pi + \frac{4}{3} \right]$$

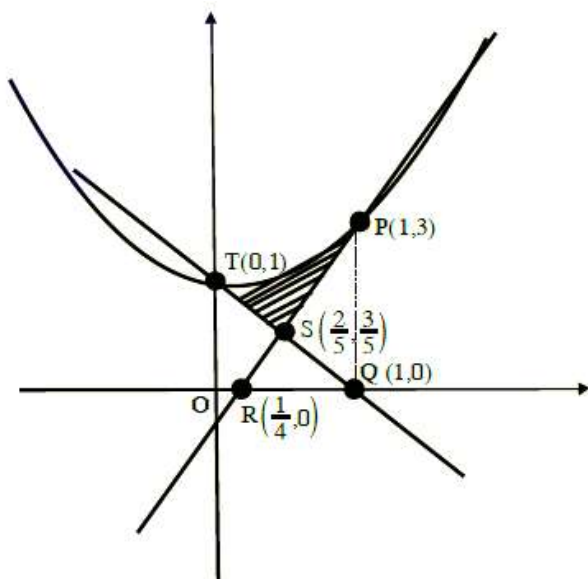
$$= 16$$

190. If A is the area in the first quadrant enclosed by the curve C : $2x^2 - y + 1 = 0$, the tangent to C at the point (1, 3) and the line $x + y = 1$, then the value of 60A is _____.

यदि वक्र C : $2x^2 - y + 1 = 0$, C के बिंदु (1, 3) पर स्पर्श रेखा तथा रेखा $x + y = 1$ से घिरे, प्रथम चतुर्थांश में, क्षेत्र का क्षेत्रफल A है, तो 60A बराबर है _____

Ans. 16

Sol.



$$y = 2x^2 + 1$$

Tangent at (1, 3)

$$y = 4x - 1$$

$$A = \int_0^1 (2x^2 + 1) dx - \text{area of } (\Delta QOT) - \text{area of } (\Delta PQR) + \text{area of } (\Delta QRS)$$

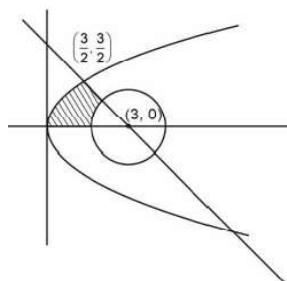
$$A = \left(\frac{2}{3} + 1 \right) - \frac{1}{2} - \frac{9}{8} + \frac{9}{40} = \frac{16}{60}$$

191. If the area bounded by the curve $2y^2 = 3x$, lines $x + y = 3$, $y = 0$ and outside the circle $(x - 3)^2 + y^2 = 2$ is A, then $4(\pi + 4A)$ is equal to _____.

यदि वक्र $2y^2 = 3x$, रेखाओं $x + y = 3$, $y = 0$ तथा वृत्त $(x - 3)^2 + y^2 = 2$ के बाहर गिरा हुआ क्षेत्रफल A है, तो $4(\pi + 4A)$ मान होगा _____

Ans. 42

Sol.



$$\int_0^{3/2} \left((3 - y) - \frac{2y^2}{3} \right) dy - \pi (\sqrt{2})^2 \cdot \frac{1}{8}$$

$$= \frac{36-9-6}{8} - \frac{\pi}{4} = \frac{21}{8} - \frac{\pi}{4}$$

$$4(\pi + 4A) = 42$$